Tab 1

• 8 Queens problem

```
]: N = 8

def print_solution(board):
    for row in board:
        print(" ".join("Q" if col else "." for col in row))
    print("\n")


def is_safe(board, row, col):
    for i in range(row):
        if board[i][col] or \
           (col - (row - i) >= 0 and board[i][col - (row - i)]) or \
           (col + (row - i) < N and board[i][col + (row - i)]):
            return False
    return True


def solve_n_queens(board, row):
    if row == N:
        print_solution(board)
        return True

    found = False

    for col in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            found = solve_n_queens(board, row + 1) or found
            board[row][col] = 0

    return found


board = [[0] * N for _ in range(N)]

solve_n_queens(board, 0)
```

Tab 2

```
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

```
Q . . . . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
. . . . . . Q .
. . . Q . . . .
. Q . . . . . .
. . . . Q . . .
```